



S H E N T O N
C O L L E G E

2017 ATAR CHEMISTRY

UNITS 1 & 2

TEST 1
ANSWERS

**Properties and Structure of Atoms
and
Bonding**

MULTIPLE CHOICE SECTION

DO NOT WRITE ON THIS BOOKLET

PLEASE DO NOT TURN THE PAGE UNTIL INSTRUCTED TO DO SO

Total Recommended time: 50 minutes

PART A: MULTIPLE CHOICE SECTION

Place a cross (X) through your answer on your answer booklet. DO NOT write in this booklet.

1. A sulfate ion (SO_4^{2-}) contains the isotopes S-33 and O-15. How many electrons, protons and neutrons does the ion possess?

	electrons	protons	neutrons
a.	46	48	48
b.	93	91	96
c.	50	48	45
d.	48	45	50

2. An unknown solid material is added to water and stirred. On filtration a residue remains on the filter paper. Further experiments show that the quantity of residue is unaffected by addition of more water. When the filtrate (the solution that went through the filter paper) is evaporated, a solid residue is found in the container. From this information the original unknown material must have been:

- a. **a mixture**
- b. an element
- c. a pure substance
- d. a compound

3. Which of the following formulae represents a molecule of a compound?

- a. S_8
- b. NO_2**
- c. MgCl_2
- d. Si

4. The ion X^{3+} has an electronic configuration 2,8. Therefore element X is in:

- a. Group 2, Period 3
- b. Group 13, Period 2
- c. Group 13, Period 3**
- d. Group 14, Period 2

5. Which of the following would conduct an electric current?

- I a crystal of sodium chloride
- II molten silver nitrate
- III an aqueous solution of ammonium sulfate
- IV magnesium

- a. I, II and III only
- b. II, III and IV only**
- c. II and IV only
- d. II and III only

For questions 6 – 8 use the following representation of a Periodic Table.

I	II							III	IV	V	VI	VII	VIII
													K
J	A							L	F	M		E	
								B	D				
												H	G

Elements are NOT listed by their usual symbols.

6. Which of the following pairs of elements is least likely to form compounds?
 - a. A and E
 - b. M and F
 - c. B and D
 - d. G and K

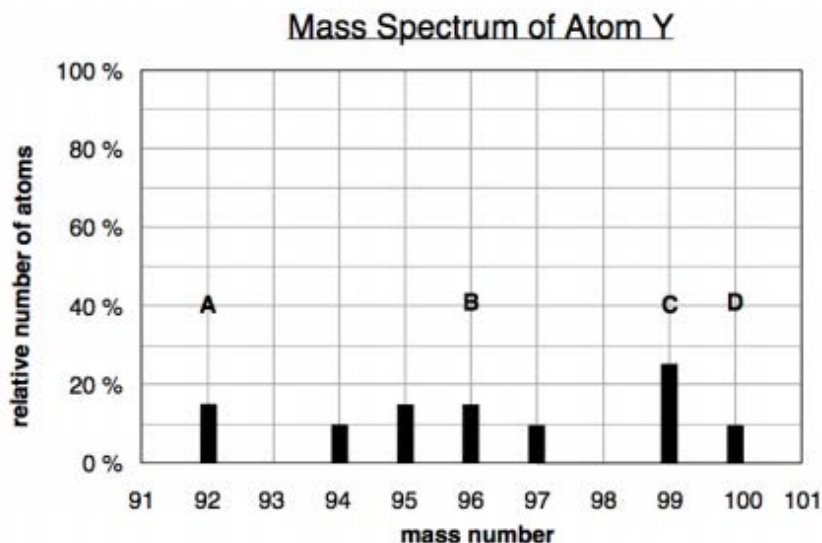
7. Which of the following elements could have a melting temperature of 660°C and be a good conductor of electricity as a solid?
 - a. E
 - b. L
 - c. G
 - d. F

8. Which of the following pairs of elements would form a covalent molecular compound?
 - a. J and A
 - b. D and L
 - c. E and K
 - d. M and E

9. Which of the following is INCORRECT?
 - a. Mixtures cannot be separated by physical means.
 - b. Elements and compounds are pure substances.
 - c. Sublimation refers to the phase change of a solid to a gas.
 - d. Mixtures can be homogeneous or heterogeneous.

10. Listed below are the atomic numbers of different elements. Which of the sets of elements contains atoms with similar properties?
 - a) 1, 2 and 3
 - b) 1, 3 and 11
 - c) 7, 16 and 35
 - d) 13, 31 and 49

11. Based on the mass spectrum of atom Y below, which of the following statements is FALSE?



- a. Peak A and Peak D come from atoms that have the same number of electrons
- b. There are 7 isotopes of atom Y
- c. Peak C comes from the most abundant isotope of atom Y
- d. Peak D comes from an atom with 4 more protons than the atom that gave Peak B

12. Which of the following lists the steps in mass spectrometry in the correct order?

- a. Detection, ionisation, acceleration, deflection
- b. Ionisation, deflection, acceleration, detection
- c. Ionisation, acceleration, deflection, detection
- d. Acceleration, ionisation, deflection, detection

13. Which of the following statements regarding nanotechnology is TRUE?

- a. Nanomaterials have different physical properties to the bulk material
- b. Objects on a nanoscale can still be seen with the naked eye
- c. Nanomaterials are materials that have particles between 100 and 1000 nm in size
- d. Nanotechnology utilises the fact that atoms in the centre of a material are more reactive than atoms on the surface of a material.

14. Which of the following mixtures could best be separated using simple (not fractional) distillation to obtain both components?

- a. Crude oil and diesel
- b. Sand and water
- c. Sugar and sand
- d. Salt solution

15. Copper wire may be bent easily without breaking. The best explanation of this is that:

- a. Slight changes in relative positions of adjacent copper particles do not break the metallic bonds as these are equally strong in all directions.
- b. The forces between copper atoms are weak, allowing copper particles to be easily moved around.
- c. Copper particles are strongly bonded in layers but only weakly bonded between layers.
- d. Copper particles are arranged in flat molecules that freely slide over each other, allowing material to be bent.

END OF SECTION A

NAME: MARKING KEY

Teacher: _____

Multiple Choice Booklet Number: _____



SHENTON
C O L L E G E

A1/A2 CHEMISTRY - TEST 1 (2017)

TIME: 50 Minutes

Properties and Structure of Atoms and Bonding

PUT A CROSS (X) THROUGH THE CORRECT ANSWER.

1.	a	b	c	d
2.	a	b	c	d
3.	a	b	c	d
4.	a	b	c	d
5.	a	b	c	d
6.	a	b	c	d
7.	a	b	c	d
8.	a	b	c	d

9.	a	b	c	d
10.	a	b	c	d
11.	a	b	c	d
12.	a	b	c	d
13.	a	b	c	d
14.	a	b	c	d
15.	a	b	c	d

PART A /15	
PART B /35	
TOTAL /50	
PERCENTAGE	

PART B: WRITTEN SECTION

1. Calculate the number of protons, neutrons and electrons in each of the following; the number represents the mass number of this particle.

Atom/ion	Number of protons	Number of electrons	Number of neutrons
Sodium-23 ion	11	10	12
Arsenic-74 Atom	33	33	41
Oxide-18 ion	8	10	10
Neon-20 atom	10	10	10

-1/2 mark for each incorrect box to a max of 4 marks lost (4 marks)

2. Classify the following substances as elements, compounds or mixtures. If it is a mixture state whether it is homogeneous or heterogeneous.

SUBSTANCE	ELEMENT, COMPOUND, MIXTURE	HOMOGENEOUS / HETEROGENEOUS
Solid copper oxide	Compound	
A cordial solution	Mixture	Homogeneous
Chocolate chip biscuit	Mixture	Heterogeneous
Nitrogen gas	Element	

1/2 mark per box (3 marks)

3. Use the following information to answer the questions below about atoms of elements A, B, C, D and E. **(5 marks)**

ELEMENT	ATOMIC NUMBER	MASS NUMBER	ELECTRONIC STRUCTURE
A	3	7	2,1
B	17	36	2,8,7
C	10	21	2,8
D	12	24	2,8,2
E	19	39	2,8,8,1

- a. Which element has 20 neutrons in each atom? **E** (1)
- b. Which two elements are in the same Group of the Periodic Table?
 A & **E** (1)
- c. Which two elements **form ions** with the same electronic structure as argon?
 B & **E** (1)
- d. The above mass number of Element D is not rounded off and the relative atomic mass is listed as 24.3 on the Periodic Table. Why is the relative atomic mass different to the mass number in this question? (2)

The relative atomic mass takes the abundance of all isotopes into account and provides an average (1),

whereas the mass number shows the relative atomic mass for a single atom/isotope only (1).

OR

the mass number is the total number of protons and neutrons in the nucleus of a particular atom

4. Calculate the relative atomic mass of a sample of lithium, given that a mass spectrometer shows that it consists of 7.40% of ${}^6\text{Li}$ and 92.6% of ${}^7\text{Li}$. Show all your working. **(3 marks)**

(% abundance x A_r of isotope #1) + (% abundance x A_r of isotope #2) (1)

(7.4/100 x 6) + (92.6/100 x 7) (1)

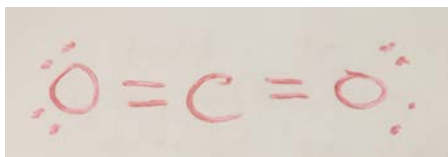
= 6.93 (3 sf) (1)

- ½ mark if incorrect number of sig. figs. or if units included

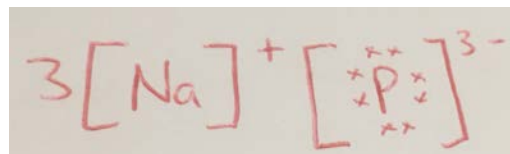
5. Draw electron dot diagrams of the following substances.

(8 marks)

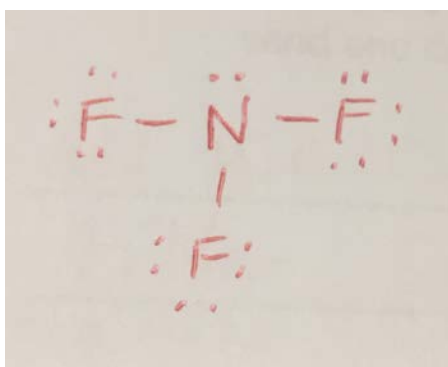
a. CO_2



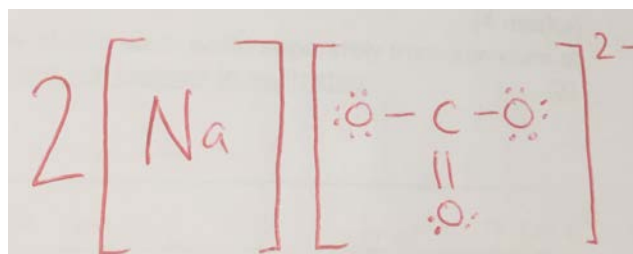
b. Na_3P



c. NF_3



d) Na_2CO_3



2 marks/diagram. No marks if incorrect type drawn (ionic/CM). -1 per mistake
Lines okay as bonds instead of dots. Must have square brackets to indicate ionic bonding.

6. Suggest a property that could be used to distinguish between each of the following different types of solid. In each case, give a detailed explanation for your choice, linking the property to the structure of the materials. (6 marks)

a. An ionic solid and a metal.

Property

Electrical conductivity (as a solid)

(1)

Reason

Ionic solids will not conduct electricity whereas a metal will.

(1)

This is because ionic solids have ions that are immobile whereas metals contain delocalised electrons that can carry the charge.

(1)

- b. A covalent network solid and a molecular solid.

Property

Melting point **or** **hardness** **(1)**

Reason

Covalent network substances have very high MP whereas covalent molecular substances have a low MP **(1)**

This is because it takes a large amount of energy to break many strong covalent bonds between atoms to melt a covalent network substance **(½)**
but only a small amount of energy to break weak intermolecular forces of attraction to melt a covalent molecular substance. **(½)**

OR

Covalent network substances are very hard whereas covalent molecular substances are soft **(1)**

This is because it takes a large amount of energy to break many strong covalent bonds between atoms in a covalent network substance **(½)**
but only a small amount of energy to break weak intermolecular forces of attraction in a covalent molecular substance. **(½)**

7. Give two reasons why atomic absorption spectroscopy (AAS) might be used instead of a flame test.

- I) **It is quantitative rather than qualitative** **(1)**
II) **Colours of similar wavelengths from two different metals may look identical to the naked eye in flame tests** **(1)**

8. **(4 marks)**
a. Outline the steps you would take to obtain each **solid** separately from a mixture of sand and copper sulfate. Include what will happen in each step. **(3)**

1: Dissolve the mixture in water (½) – copper sulfate dissolves, sand doesn't (½)

2: Filter out sand from the solution (½)– solution passes through, sand doesn't (½)

3: Crystallise copper sulfate (½)– water evaporates, leaving copper sulfate behind (½)

- b. Which physical properties (at least 2) are being used to separate the mixture? **(1)**

Solubility of solid state

Size of particle

Boiling point

½ mark each for any 2.

Written Section Total

/35